

Weight Manager · Zero-Basis Tutorial (English)

Weight Manager · v1.8.0 · 2026-08-14 · Compatible with Blender 4.2+ / 5.x

Written for users who have never painted weights before. Every concept is explained in plain language first, then shown exactly where to click.
Order: understand the concept → install → first run → feature by feature → real workflow → troubleshooting.

Step 0: What "weight" means in one minute

A 3D character moves because bones drive the skin:

bones mesh (the "skin") | shoulder | | a
shell of small | | elbow | =====> | triangles made of | | wrist | assignment | many vertices |

The model is a skin made of many vertices (little points on the mesh). A vertex can't move by itself — it has to obey the bones. Weight is the assignment table that says "this vertex listens to which bones, and how much to each."

- Weights are numbers from 0.0 to 1.0. 0 = this bone has no control; 1 = this bone fully controls the point.
- One point can be controlled by several bones, e.g. the point on the inside of the elbow = upper-arm bone 60% + forearm bone 40%. That's how a bent elbow keeps the skin from tearing.
- For every vertex, all its weights add up to 1 (this is called normalizing). The example above is $0.6 + 0.4 = 1$.

A Vertex Group in Blender is "the list of points that one bone controls, plus how much it controls each." So the "vertex group list" in the panel = the bone list. Editing weights is really editing this assignment table.

One-liner: painting weights = deciding "which bone each mesh point belongs to, and how strongly." Do it well and the character deforms naturally; do it badly and the skin tears or twists when the character moves.

Step 1: Installation

There are two ways — classic addon and official extension. Either one works.

Option A: Classic addon (zip)

1. Open Blender → menu Edit → Preferences → Add-ons
2. Click Install... (top right), pick weight_manager.zip
3. Type weight manager in the search box, tick "Mesh: Weight Manager"

Option B: Official extension (recommended, Blender 4.2+)

1. Open Blender → Edit → Preferences → Get Extensions (or in Add-ons, switch the dropdown to Extensions)
2. Click Install from Disk..., pick the extension zip (the one whose root has blender_manifest.toml)
3. Tick Weight Manager

After installing: press N in the 3D viewport to open the right sidebar — a new Weight Mgr tab will be there. That's where this addon lives.

Step 2: Before you can use the panel

The panel only shows for mesh objects, and most features need Edit Mode or Weight Paint Mode.

1. Select a mesh object (e.g. the default cube)
 2. The Weight Mgr tab appears — click it
 3. The "Vertex Groups (bones)" list is empty — that's normal. Let's make some "bones".
-

Step 3: First run (5 steps and you're in)

We'll use the default cube.

① Make two "bones"

In the panel under "Vertex Groups (bones)", click New, name it e.g. Bone_Left; create another named Bone_Right.
(On a real character these vertex groups are auto-generated by "Automatic Weights" when you parent the mesh to an armature — you don't hand-make them. This is just for learning.)

② Enter Weight Paint Mode

Press Ctrl+Tab and pick Weight Paint from the menu.
Or use the mode dropdown (top-left of the viewport) → Weight Paint.

③ Assign points to a bone

- The current group is Bone_Left. Hover the mouse over the mesh, hold left mouse and drag to paint — painted areas glow orange/yellow, meaning "these points now belong to Bone_Left".
- Want the right side too? Turn on Symmetry X in the panel, or paint the left side and use the mirror command later.

④ Set exact values with a slider (this addon's core, like C4D)

1. Enter Edit Mode (Tab)
2. Box-select (B) some vertices, or press A to select all
3. In "Auto Weight" choose Absolute (the leftmost mode)
4. Drag the big slider to 50% — every selected point instantly becomes 0.50, and the number updates live next to it.

That's the core C4D-style interaction: no brushing back and forth — set exact numbers with a slider.

⑤ Look at the per-point numbers

1. In Edit Mode select 1–3 vertices
 2. The "Vertex Weight Table" appears at the bottom of the panel
 3. Top list = selected vertices (click to switch); bottom list shows every bone's weight on that vertex — drag any row's slider to edit it directly.
-

Step 4: Every section of the panel (top to bottom)

① Vertex Groups (bones) list — who is the current bone

- Clicking a row makes it the current group; everything else operates on it.
- Each row has a small percentage bar on the right: the average weight of the currently selected vertices on this bone — a quick sense of how much this bone covers.
- The lock icon ([lock]) at the end of a row: click to lock. Locked groups are read-only — Normalize will not touch them (like C4D's "lock joints during normalize").
- Buttons below: New / Delete / Rename.

② Influence Highlight — which points a bone covers

- "Highlight current group influence" (on by default). In Edit/Weight Paint mode, pick a bone and the viewport marks the points it influences (weight > 0) with orange dots.
- Look before you paint — avoid "this bone reaching points it shouldn't."

③ Joint Filter — too many bones to see clearly

- Tick "Only show joints that affect the selection": the list keeps only bones that have weight on the selected points; unrelated bones are hidden.
- "Search bone name" box filters by name. The two combine.

④ Weight HUD — value follows your cursor

- Tick "Weight HUD". In Edit/Weight Paint mode, move the mouse over the mesh and the cursor shows bone name: weight in real time.
- Like C4D's viewport HUD — no need to look down at the panel while painting.

⑤ Select Mask + Symmetry (top of Weight Paint Mode)

- Face mask / Vertex mask: in Weight Paint mode Blender hides the wireframe and you can't box-select points by default. These two toggles expose it. With vertex mask on, unselected vertices show black (normal) — box-select first, then paint; if the black is annoying use face mask instead (unselected faces just dim).
- Shortcuts: Alt+Left select face, Shift+Alt add, B box select, A select all.
- Symmetry X/Y/Z: when on, painting the left side also paints the right (like C4D symmetric painting).

⑥ Auto Weight — the core painting area

Pick a mode, then drag the big slider (or hover over the number and press Ctrl+wheel):

- Absolute: wherever the slider is, the selected points' weight becomes that value instantly.

- Add / Subtract: drag as far as you like — it adds/subtracts by how much you dragged, regardless of direction. The slider resets to zero after each gesture, so you can drag repeatedly and keep accumulating.

- Smooth: the more you drag, the closer the selected weights move toward the average of their neighbors; Radius (1–5) = how far the neighborhood reaches (bigger radius, smoother over a wider area).

- "Apply once (by Strength)": don't want to drag? Set Strength and click once.

Tip: Ctrl+wheel over a value is native Blender fine-tuning. A bare wheel doing nothing is normal.

⑦ Commands — undo, mirror, copy/paste

- Invert: weight $\rightarrow 1 - \text{weight}$ (white becomes black and vice versa — great for "paint the parts I missed").
- Mirror: copy the weights of selected points to the symmetric position (pick X/Y/Z axis). Double-check the axis if your model isn't perfectly symmetric.
- Copy Weights / Paste Weights (like C4D):
 1. Select a set of points, press Copy Weights (stores the current bone's values on those points, in selection order)
 2. Select a different set of points (or a different bone), press Paste Weights
 3. Values are pasted one-to-one in selection order; if the counts differ, only the first few are pasted (with a note).

⑧ Normalize — make everything sum to 1

- Select points $\rightarrow$ press "Normalize selection (sum = 1)".
- Locked groups ([lock]) keep their weights; only the other groups are rescaled to make up the difference.
- When to use: you painted past 1 by hand, or Absolute made some points sum > 1 and the animation jumps. Normalize once and it's fixed.

⑨ Select by Weight — grab the points you want fast

- =0 (no weight), >0 (some), <1 (not fully), =1 (fully owned), ≈range (near a threshold).
- E.g. find "points I forgot to paint" → press =0.

⑩ Vertex Weight Table — read/edit per-point numbers like a spreadsheet

- Appears when vertices are selected in Edit/Weight Paint mode.
- Top list: selected vertices (click to switch); bottom list: every bone's weight on that vertex, sorted descending, zeros at the bottom.
- Dragging a row's slider = editing that vertex's weight on that bone directly. Dragging a 0 up = adding this bone to that point.
- Locked bones' rows are greyed out and can't be dragged.

Fill Select — works with the Fill Select addon

- Requires the "Fill Select" addon (bl_ext.fill_select_between).
 - Usage: select two "edge loops" as boundaries → click "Fill Select" → the faces between the loops get selected → paint weights on them.
 - Perfect for painting a belt or a cuff evenly.
-

Step 5: A complete real workflow (painting a character arm)

1. Armature-parent and auto-weight (Ctrl+P → With Automatic Weights). Blender now auto-creates vertex groups per bone.
 2. Enter Weight Paint Mode, pick a bone in the panel list, look at the orange highlight: which points does this bone own, and is it right?
 3. Fine-tune: Edit Mode → box-select the area → in Auto Weight use the Absolute slider to drag the value to the percentage you want.
 4. Deal with the joint: select the loop of points inside the elbow, confirm "upper arm + forearm half-and-half" (check both rows in the Vertex Weight Table, drag each to ~0.5).
 5. A bone owns too much: select the wrong points → Absolute → drag the value to 0 (or select by =0 and zero them).
 6. Left = right: after painting the left side, use Commands → Mirror to copy it.
 7. Sum ≠ 1 / animation jumps: select all points → Normalize.
 8. Play the animation; where it tears, go back to step 3.
-

Step 6: FAQ / troubleshooting

Q: The panel is greyed out / buttons won't press.

A: You need Edit Mode or Weight Paint Mode (Tab or Ctrl+Tab) and a mesh object selected.

Q: Everything is black in Weight Paint mode.

A: That's vertex mask on — normal. Unselected vertices show black. Box-select (B) or Alt+click vertices first; only the selection shows color. If the black is annoying, switch to face mask.

Q: Spinning the wheel over the number does nothing.

A: You need Ctrl+wheel (hover over the value/slider). A bare wheel is native Blender behavior and does nothing.

Q: No bones in the list.

A: There are no vertex groups yet. Click New in the "Vertex Groups" area; or Ctrl+P → Automatic Weights first — that creates the bones' vertex groups for you.

Q: Can't paint / weights don't change.

A: Check if the group is locked ([lock]). Locked = read-only — unlock first. Also check that the bone you selected is the one you want to paint.

Q: The Vertex Weight Table doesn't show.

A: You need Edit/Weight Paint mode and selected vertices. Both must be true.

Q: Smooth radius seems to do nothing.

A: Radius 1 = only direct neighbors. Want a bigger sweep? Set radius 2/3 and drag again.

Q: The addon disappeared after switching Blender versions.

A: Every Blender version uses its own addon directory. Copy the weight_manager folder to the new version's scripts/addons (or re-Install the zip there). Same for the extension: install the zip again from the new version's "Install from Disk".

Appendix: How this maps to C4D / Maya

This addon	Counterpart
Vertex Groups (bones) list	C4D Weight Manager's Joints list
Auto Weight sliders	C4D Weight Manager's Auto Weight
Copy / Paste weights	C4D's Copy / Paste
Normalize + locked groups	C4D's lock-joints normalize
Influence highlight, Weight HUD	C4D joint influence view / weight HUD
Vertex Weight Table (per-point numbers, editable)	Maya's Component Editor (Excel-like)
Select by weight	C4D's Fill Selection

One-liner: the "panel + slider" painting experience of C4D, plus the "read/edit per-point numbers" precision of Maya, packaged as a Blender sidebar addon.